

MySQL Data Types — D426 Quick Reference

For OA study and Zybooks labs

Signed vs Unsigned: `ColumnName INT = signed (default, allows negatives)` | `ColumnName INT UNSIGNED = unsigned (positive only)`

Integer Types

Type	Bytes	Signed Range	Unsigned Range	Good For
TINYINT	1	-128 to 127	0 to 255	Flags, small counters, boolean-like
SMALLINT	2	-32,768 to 32,767	0 to 65,535	Department codes, limited-range IDs
MEDIUMINT	3	-8,388,608 to 8,388,607	0 to 16,777,215	Medium-range IDs
INT / INTEGER	4	-2,147,483,648 to 2,147,483,647	0 to 4,294,967,295	Most common — general IDs, counts
BIGINT	8	⁶³ -2 to ⁶³ 2 - 1	⁶⁴ 0 to ⁶⁴ 2 - 1	Very large IDs, social media scale

OA Tip: Match the question's range to the smallest type. "Positive integer max 65,000" → **SMALLINT UNSIGNED**.
"Max 255" → **TINYINT UNSIGNED**.

Decimal / Numeric Types

Type	Bytes	Precision	Syntax	Good For
DECIMAL (M,D)	Varies	Exact — M total digits, D after decimal	<code>DECIMAL (8, 2)</code> = 999,999.99	Money, prices, financial — exact values needed
FLOAT	4	Approximate — ~7 significant digits	<code>FLOAT (no params)</code>	Measurements, weight — slight rounding OK
DOUBLE	8	Approximate — ~15 significant digits	<code>DOUBLE (no params)</code>	Higher-precision scientific calculations

OA Tip: `DECIMAL(M,D)` — M = total digits, D = digits after decimal. So `DECIMAL(8,2)` = 6 digits before + 2 after = 999,999.99. "Variable precision, 4 bytes" → **FLOAT**.

String Types

Type	Length Behavior	Syntax	Good For
CHAR (N)	Fixed — always N characters, padded with spaces if shorter	<code>CHAR (3)</code>	State codes, country codes, fixed-format codes — always same length
VARCHAR (N)	Variable — up to N characters, no padding	<code>VARCHAR (50)</code>	Names, titles, descriptions — varies in length

OA Tip: Watch for "fixed-length" → CHAR vs "variable-length" → VARCHAR. Both use single quotes when inserting: 'like this'

Date & Time Types

Type	Stores	Format	Good For
DATE	Year, month, day	<code>'YYYY-MM-DD'</code>	Birthdays, release dates, hire dates
TIME	Hours, minutes, seconds	<code>'HH:MM:SS'</code>	Durations, shift times
DATETIME	Year, month, day + hours, minutes, seconds	<code>'YYYY-MM-DD HH:MM:SS'</code>	Timestamps, log entries, orders
TIMESTAMP	Same as DATETIME but stored as UTC	<code>'YYYY-MM-DD HH:MM:SS'</code>	Auto-tracked creation/modification times

OA Tip: "Year, month, day" → DATE. "Year, month, day and time" → DATETIME. Neither takes parameters. Date values use single quotes: '2020-01-15'

DECIMAL(M,D) Quick Examples

Requirement	M (total digits)	D (after decimal)	Syntax	Max Value
Up to \$999,999.99	8	2	<code>DECIMAL (8, 2)</code>	999,999.99
6 sig digits, 1 after decimal	6	1	<code>DECIMAL (6, 1)</code>	99,999.9
Up to \$9,999.99	6	2	<code>DECIMAL (6, 2)</code>	9,999.99
Percentage with 2 decimals	5	2	<code>DECIMAL (5, 2)</code>	999.99

Remember: M = digits before decimal + digits after decimal. For \$999,999.99 that's 6 + 2 = 8, so DECIMAL(8,2).

Please note that this is not an all-encompassing guide to Data types in SQL. It's just a list that should get you through WGU's D426 Database Foundations Class as of May 2026. The further out from this date that you look at this reference sheet, the more likely it is to be missing material you may need to know. This document was created using Claude AI and is subject to having errors that I did not catch or correct.